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STATUTORY INSPECTION REPORT
(Under Factories Act 1948 & PESO Regulations)

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Inspection Agency : Maharashtra Industrial Safety and Health (MISH)
Inspector Name : Mr. P.V. Kulkarni (Inspector of Factories)
Inspector ID : MISH-INS-0442
Facility : Bharat Process Industries Pvt. Ltd.
Facility Address : Plot 44, MIDC Taloja, Navi Mumbai - 410208
Date of Inspection : 15-January-2024
Report Date : 22-January-2024
Report Reference : MISH-IR-2024-BPIL-007

SECTION A – FACILITY INFORMATION

Factory License No : FACT-MH-2016-00892
Factory Manager : Mr. Anil Gupta
Contact Person : Mr. Vikram Reddy (Sr. Maintenance Manager)
Units Inspected : Unit 3 – Cooling Water and Steam System
Number of Workers : 124 (total plant), 18 (Unit 3)
Previous Inspection : 14-July-2023 (Report Ref: MISH-IR-2023-BPIL-019)

SECTION B – EQUIPMENT INSPECTED

B.1 PRESSURE VESSEL – V-22 (Steam Pressure Vessel 22)

Manufacturer : BHEL
Design Pressure : 25.0 bar (g)
Working Pressure : 22.0 bar (g)
Design Temperature : 220°C
Year of Manufacture : 2017
Capacity : 8,000 litres
Last Certificate No : IBR-PV-2023-MH-00231

Inspection Findings:

- (i) External visual inspection: No corrosion, cracks, or deformations observed on shell, heads, or nozzles.
- (ii) Thickness gauging (ultrasonic): Minimum recorded thickness 18.2 mm at 12 measurement points. Design thickness: 18.5 mm. Corrosion allowance remaining: 2.7 mm. Adequate.
- (iii) Safety Relief Valve (SRV-22A): Set pressure verified at 24.8 bar – within acceptable tolerance of $\pm 2\%$ of design. Lifting tested – satisfactory. Last calibration certificate valid up to December 2024.
- (iv) Pressure gauge calibration: PG-V22 found reading 0.4 bar HIGH – calibration required. Non-compliance noted.
- (v) Nameplate: Legible and intact.
- (vi) Vessel log book: Maintained. All entries current.
- (vii) Foundation bolts: Inspected – no looseness observed.

Compliance Status : CONDITIONAL PASS

Non-Compliance : Pressure gauge PG-V22 requires recalibration within 30 days. Compliance report to be submitted to MISH by 15-February-2024.

Certificate Valid : 15-January-2024 to 14-July-2024

Certificate No : IBR-PV-2024-MH-00089

B.2 PRESSURE VESSEL – V-23 (Steam Pressure Vessel 23)

Manufacturer : BHEL
Design Pressure : 25.0 bar (g)
Working Pressure : 22.0 bar (g)
Design Temperature : 220°C
Year of Manufacture : 2017

Capacity : 8,000 litres
Last Certificate No : IBR-PV-2023-MH-00232

Inspection Findings:

- (i) External visual inspection: Minor surface rust on saddle supports – wire brushing and painting recommended.
- (ii) Thickness gauging: Minimum 18.6 mm – above design minimum.
- (iii) Safety Relief Valve (SRV-23A): Note – spring fatigue detected during this inspection. SRV replaced before certificate issue. New spring installed and set pressure calibrated at 24.5 bar. New calibration cert issued.
- (iv) Pressure gauge PG-V23: Within calibration. Valid to Sep-2024.
- (v) Manhole cover: Gasket replaced in February 2024 per plant maintenance records (WO-2024-006). Satisfactory.

Compliance Status : PASS (with recommendations)
Recommendation : Repaint saddle supports within 3 months.
Certificate Valid : 01-February-2024 to 31-July-2024
Certificate No : IBR-PV-2024-MH-00090

B.3 BOILER – BL-07 (Fire Tube Boiler)

Manufacturer : Thermax Ltd.
Design Pressure : 32.0 bar (g)
Working Pressure : 28.0 bar (g)
Design Temperature : 250°C
Year of Manufacture : 2016
Capacity : 1,000 kg/hr steam output
Last IBR Certificate : IBR-BLR-2023-MH-00156
Registration No : IBR-BLR-REG-MH-2016-00892-BL07

Inspection Findings:

- (i) Fireside inspection: Soot deposits moderate – cleaning done before inspection. Refractory – minor crack on left wall repaired by plant team. Post-repair satisfactory.
- (ii) Waterside inspection: Scale deposits within acceptable limits (< 1.5 mm). Water treatment records reviewed – chemical dosing records for Jan-2024 reviewed and adequate.
- (iii) Safety valves: Both SV-BL07-A and SV-BL07-B tested and set pressures verified. Satisfactory.
- (iv) Gauge glass: Replaced per WO-2024-010. Discolouration on old gauge glass was noted as observed finding. New glass installed – satisfactory.
- (v) Burner and fuel system: Gas valve train inspected. All solenoid valves tested. Flame failure relay tested.
- (vi) Water level controls: Low water cut-off tested – tripped correctly at set level. High water alarm tested – OK.
- (vii) Blowdown valve: Leaking slightly past seat. Maintenance notified – corrective action required within 15 days.

Compliance Status : CONDITIONAL PASS
Non-Compliance 1 : Blowdown valve (BDV-BL07) seat leakage – repair required within 15 days.
Non-Compliance 2 : IBR-registered operator log book incomplete for 3 entries in December 2023. Complete records within 7 days.
IBR Certificate : 18-March-2024 to 17-March-2025
Certificate No : IBR-BLR-2024-MH-00103

SECTION C – GENERAL FACTORY ACT COMPLIANCE OBSERVATIONS

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- C.1 First Aid facilities: Adequate. First aid boxes inspected – stocked as per Schedule 18 (Factory Act).
 - C.2 Fire extinguishers near Unit 3: 4 CO2 and 2 DCP units present. All within service dates. PASS.

- C.3 Emergency exits: 3 exits available for Unit 3. Clearly marked. Exit signs illuminated. PASS.
- C.4 Personal Protective Equipment: PPE board present. Helmets and safety shoes observed in use. PASS.
- C.5 Hazardous Chemical Storage (TK-09 area): Chemical (biocide and scale inhibitor) drums stored correctly. MSDS displayed. PASS.

SECTION D – SUMMARY OF NON-COMPLIANCES AND DEADLINES

No.	Equipment	Non-Compliance	Deadline
1	V-22	Recalibrate pressure gauge PG-V22	15-Feb-2024
2	BL-07	Repair blowdown valve BDV-BL07	01-Apr-2024
3	BL-07	Complete IBR operator log entries	25-Jan-2024

SECTION E – NEXT INSPECTION

Next statutory inspection scheduled: July 2024
Bharat Process Industries is required to submit compliance certificates for items in Section D by the deadlines stated.
Failure to comply will result in show cause notice under Section 36A of the Factories Act, 1948.

Signed:

Inspector : P.V. Kulkarni, MISH
Plant Manager : Anil Gupta, BPIL
Date : 22-January-2024